# BOTTOM-ROW QUADRATIC REDUCTION AND SCALAR-SPAN OBSTRUCTION ON RANK-FOUR PARABOLIC ERDŐS–ULAM CONFIGURATIONS

ARGUS OPEN RESEARCH PROJECT

Abstract. The Erdős–Ulam problem asks whether there exists a dense subset of $\mathbb{R}^2$ all of whose pairwise Euclidean distances are rational. In the Gaussian rank-four parabolic parameterization at scale $\lambda = 27$, we prove an exact quadratic reduction of the bottom distance row and a scalar-span rank-deficiency obstruction ruling out the rank-deficient non-star sub-branch.

## 1. Main Algebraic Reduction

**Proposition 1.1.** *On the Gaussian rank-four parabolic locus at parameter $\lambda = 27$, clearing the first three base-point rational-distance squares reduces the fourth distance condition to a binary quadratic norm equation whose scalar-span specialization has rank at most 2 and admits only collinear/star-degenerate rational solutions.*

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